

Neuro-Symbolic Dynamic Atlas: Reproducibility, Ablation, and Robustness Analysis

Automated Scientific Report — Phase 8

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Contents

1	Introduction	2
2	Reproducibility Audit (Phase 8A)	2
3	Ablation Study (Phase 8B)	3
4	SOTA Benchmark (Phase 8C)	3
5	Robustness Stress Test (Phase 8E)	3
5.1	Noise Robustness	3
5.2	Missing Data	3
5.3	Parameter Drift	3
5.4	OOD Generalization	4
6	Computational Cost	4
7	Threats to Validity	4
7.1	Statistical Reproducibility	4
7.2	Synthetic Dataset Bias	4
7.3	Hardware Dependence	4
7.4	Hyperparameter Sensitivity	4
7.5	Domain Transfer Limitations	4
7.6	Simulator-Reality Gap	4
8	Conclusions	5

Abstract

We present a systematic evaluation of the Neuro-Symbolic Dynamic Atlas pipeline across 63 module-system combinations. Reproducibility analysis (BCa bootstrap, Sobol seeds, adaptive stopping) shows 28.6(median CV = 0.0373). Ablation study identifies 0 large-impact removals ($|d| \geq 0.8$). Mean OOD gap: 0.0861. **All results are model-specific observations on numerical simulations.**

1 Introduction

Scientific machine learning pipelines require validation beyond held-out accuracy. This report covers: (1) reproducibility, (2) ablation, (3) SOTA comparison, (4) robustness.

2 Reproducibility Audit (Phase 8A)

Seeds: $S = \lfloor p \times (2^{31} - 1) \rfloor$ (Sobol) [?]. BCa bootstrap CI_{95} [?] with $n_{\text{resamples}} = 2000$. Convergence: $W_{\text{rel}} = (CI_{hi} - CI_{lo})/|\mu| < 0.05$ (max 50 seeds).

Table 1: Reproducibility Audit Summary

Metric	Value
Total evaluations	63
Converged ($W_{\text{rel}} < 0.05$)	28.6Stable (CV < 0.05)
23.8Median CV	0.0373
Peak RAM	5.6 MB
Peak VRAM	2.3 MB

Table 2: Sample Reproducibility Results (first 10)

Module	System	Seeds	CV	Converged	Stable
EV3	lorenz	4	0.3594	False	False
EV3	duffing	50	0.3140	False	False
EV3	<i>van_der_pol</i>	20	0.0226	True	True
EV3	rossler	40	0.0755	True	False
EV3	logistic	20	0.0236	True	True
EV3	ECG200	20	0.0373	True	True
EV3	ECG5000	20	0.0373	True	True
EV3_EXT	lorenz	50	0.1749	False	False
EV3_EXT	duffing	50	0.1079	False	False
EV3_EXT	<i>van_der_pol</i>	20	0.0001	True	True

3 Ablation Study (Phase 8B)

$$\Delta\% = \frac{\mu_{\text{base}} - \mu_{\text{abl}}}{|\mu_{\text{base}}|} \times 100 \quad (1)$$

$$d = \frac{\mu_{\text{base}} - \mu_{\text{abl}}}{\sigma_{\text{pooled}}} \quad (2)$$

DAG-aware cascade disabling; AR(p) fallback (DEPENDENCY_BYPASS). Large-impact removals: 0.

Table 3: Top 8 Ablations by Cohen’s d

Config	System	Module	$\Delta\%$	d	Impact
NO_TDA	lorenz	EV3	0.0	0.000	Negligible
NO_TDA	lorenz	EV3_EXT	0.0	0.000	Negligible
NO_TDA	lorenz	EV3_DEEP	0.0	0.000	Negligible
NO_TDA	lorenz	EV3_SCI	0.0	0.000	Negligible
NO_TDA	duffing	EV3	0.0	0.000	Negligible
NO_TDA	duffing	EV3_EXT	0.0	0.000	Negligible
NO_TDA	duffing	EV3_DEEP	0.0	0.000	Negligible
NO_TDA	duffing	EV3_SCI	0.0	0.000	Negligible

4 SOTA Benchmark (Phase 8C)

pip install attempted for all missing packages. Unavailable tools: NOT_EVALUATED (no mocks). OK evaluations: 14. Not evaluated: 5. Win rate (real): 0.56; Win rate (total): 0.36.

5 Robustness Stress Test (Phase 8E)

5.1 Noise Robustness

SNR $\in \{+\infty, 20, 10, 5, 0\}$ dB. NRS = $-\text{slope}(\text{norm. metric vs SNR})$. Best: **EV3**, Mean NRS: -0.3687.

5.2 Missing Data

Dropout [0%, 10%, 30%, 50%] + linear interpolation. MDT = max drop at <20% degradation. Mean MDT: 20.0

5.3 Parameter Drift

$\sigma : 10 \rightarrow 14$ (Lorenz), $\gamma : 0.3 \rightarrow 0.5$ (Duffing).

5.4 OOD Generalization

Train: {Lorenz, Duffing}, Test: {Rössler, Van der Pol}. $GG = |\mu_{\text{in}} - \mu_{\text{ood}}|/|\mu_{\text{in}}|$. Mean gap: 0.0861.

6 Computational Cost

Table 4: Mean Computational Cost per Module (from Phase 8A)

Module	Mean Time (s)	Peak RAM (MB)	Peak VRAM (MB)
EV3	N/A	N/A	N/A
EV3_DEEP	N/A	N/A	N/A
EV3_EXT	N/A	N/A	N/A
EV3_SCI	34.01	3.8	2.1
Koopman	0.27	0.0	0.0
NeuralODE	0.01	0.0	0.0
PINN	0.01	0.0	0.0
SINDy	0.29	0.0	0.0

7 Threats to Validity

7.1 Statistical Reproducibility

BCa bootstrap validity requires non-degenerate distributions and $n_{\text{seeds}} \geq 2$.

7.2 Synthetic Dataset Bias

All systems numerically integrated; real sensor noise unmodeled.

7.3 Hardware Dependence

Timings depend on CPU/GPU model; cross-platform comparisons invalid without normalization.

7.4 Hyperparameter Sensitivity

SINDy threshold, PINN depth, NeuralODE step-size not exhaustively tuned.

7.5 Domain Transfer Limitations

1-D signal proxies of 3-D systems; multi-dimensional performance may differ.

7.6 Simulator-Reality Gap

RK4 integration only; no empirical validation. Results are model-specific observations.

8 Conclusions

Model-specific observations only. All reported metrics derive from numerical simulations. No claims about universal physical laws or out-of-domain performance are made.